

IOANNIS TSAGKAROPOULOS

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Education

M.Sc. in Computer Science, ETH Zürich

Sep 2025 – Exp. Feb 2027

- Major: Secure and Reliable Systems
- Minor: Data Management Systems
- GPA: 5.0/6

B.Sc. in Mathematics, National and Kapodistrian University of Athens

Sep 2021 – Jul 2025

- Concentration: Applied Mathematics
- GPA: 9.46/10

Apolytirio Genikoy Lykeioly, 1st Senior High School of Corfu

Sep 2018 – Jun 2021

- Graduation Grade: 20/20
- National University Entrance Examination: 19.460/20.000 (**top 0.01% nationwide**)

Work Experience

Software Engineering Intern, Blueground ([link](#))

Jan 2025 – Jul 2025

- Reduced partner integration time from **days to less than 10 minutes** by engineering an **automated integration pipeline** in **Typescript** during a major architecture migration from a legacy PHP monolith to a microservices event-driven architecture
- **Owned 2 successful E2E partner API certifications** (Hostfully, Mews) and advanced 3 additional ones by designing the system architecture, implementing E2E testing, and leading certification calls

Data Scientist Apprenticeship, SphearsAI

Jan 2023 – Dec 2023

- Enabled data-driven **retail traffic forecasting** by developing a predictive application in **R** that tracked transaction logistics, analyzed KPIs, and leveraged customer loyalty program data

Publications

Conferences

- [1] Claudio Estatico, Christos Koukouvinos, Marilena Mitrouli, and Ioannis Tsagkaropoulos. “[Landweber Banach Regression in Statistical Learning](#)”. *14th International Conference on Dependable Systems, Services and Technologies (DESSERT)*. Oct. 11, 2024–Oct. 13, 2025.

Presentations & Posters

- [1] Marilena Mitrouli, Christoforos Spyropoulos, and Ioannis Tsagkaropoulos. “[Landweber Banach Regression](#)”. Numerical Analysis and Applications to Data Science (N2ADS). Apr. 2025.
- [2] Ioannis Tsagkaropoulos. “Algorithmical Implementation of the Landweber Regression”. Numerical Analysis and Applications to Data Science (N2ADS). Apr. 2024.
- [3] Nikolaos Koukoudakis, Emmanouil Lardas, Marilena Mitrouli, Efterpi Psitou Athanasia, Markos Theocharis Kremmydas, and Ioannis Tsagkaropoulos. “[Julia Programming Language: Guide and Applications](#)”. Numerical Analysis and Scientific Computation with Applications (NASCA). July 2023.

Academic Contributions

- [1] Antonios Kominatos Gennatas and Ioannis Tsagkaropoulos. *A Guide to Computer Graphics*. Athens, Sept. 2025. URL: <https://github.com/GiannisTsagkaropoulos/Computer-Graphics-Notes>.
- [2] Marilena Mitrouli. *Arithmetical Linear Algebra and Scientific Computations*. Athens, June 2021. URL: https://hub.uoa.gr/wp-content/uploads/2023/01/mdimop_mitrouli_book.pdf.

Projects

Chacha20-Poly1305 Optimization ([link](#))

Spring 2026

- **Outperformed OpenSSL** in ChaCha20 encryption for plaintexts under 8 KiB and maintained **80% throughput** for larger inputs by implementing **C** optimizations (loop unrolling, instruction reordering, cache optimization, and ILP) specifically for the **x86** Kaby Lake microarchitecture.

Volunteering

Developer & Organiser, TEDxNTUA ([link](#)) Oct 2023 – Sep 2025

- Co-organized largest TEDxNTUA’s event to date (**900+ participants**)
- Developed 2024’s, 2025’s and the official website using **Next.js**, **React.js**, **Tailwind CSS**

Developer, Julia Team NKUA ([link](#)) Jan 2022 – Jul 2025

- Made the **Julia** Programming Language the **most used language in the Mathematics Department** via events, guides and interactive notebooks

Teaching Assistant, National and Kapodistrian University of Athens Sep 2022 – Jan 2025

- Taught class of 30 students about core programming concepts in **MATLAB** and **L^AT_EX**

Awards

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| [1] 53rd Scholarship Programme for Postgraduate Studies , Bodossaki Foundation | Jul 2025 |
| [2] Scholarship for Postgraduate Studies , Golden Union Shipping Company S.A. | Jun 2025 |
| [3] Best Conferencee Award , 7th Simulation of the ECB Conference, Get Involved | Nov 2023 |
| [4] Scholarship for Academic Excellence , Eurobank | Jun 2021 |

Technical Skills

Languages: C, Python, Typescript/Javascript, OCaml, Matlab, Julia, SQL, Assembly (x86)

Tools: Git, Github Actions, Docker, Jest, Redis, Apache Kafka, RabbitMQ, MapReduce, Spark, Hadoop

Languages

English (C2)	German (A1)	Greek (Native)
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